

Graphs Recap

Average to High Minimum Degree Every non-empty graph G has a subgraph $H \subset G$ such that $\delta(H) \geq \frac{1}{2}\bar{d}(G)$.

Linear Programming

Linear Program Maximise/Minimise $c^T x$, subject to linear constraints such as $Ax \leq b, Ax = b, x \geq 0$ where x is a vector.

Feasible Solution A vector x which satisfies all constraints. The set of feasible solutions is the feasible region, which is a convex polygon.

Optimal Solution If we want to maximise $c^T x$, then x^* is optimal if $c^T x^* \geq c^T x$ for all feasible x . If the feasible region is non-empty, either an optimal solution exists or the objective is unbounded above.

Objective Hyperplanes $c^T x$ gives a family of parallel hyperplanes $c^T x = \alpha$.

Unboundedness and Infeasibility An LP may be infeasible: no feasible x exists; unbounded: objective tends to infinity; or have at least one optimal solution.

LP Forms Inequality Form: $Ax \leq b, x \geq 0$; Standard (Equality) Form: $Ax = b, x \geq 0$.

Slack Variables Convert inequalities to equalities; for $Ax \leq b, x \geq 0$, convert to $Ax + s = b, x \geq 0, s \geq 0$. Augment A' with I , and x' with s to get $A'x' = b, x' \geq 0$.

Squeeze Equality to inequality; $Ax = b$ becomes $Ax \leq b, -Ax \leq -b$.

Farkas Lemma Applies to both equality and inequality versions; Exactly one of the following holds:

- $\exists x \geq 0, Ax = b$ or $Ax \leq b$
- $\exists y, y^T A \geq 0, y^T b < 0$ (certificate for infeasibility)

Optimality Criterion For a bounded LP, an optimal solution occurs at an extreme point (not on a line segment) of the convex feasible region. Then x^* optimal $\Rightarrow c^T x^* \geq c^T x$ for all corners x .

Duality

Primal LP $\max c^T x, Ax \leq b, x \geq 0$.

Dual LP $\min b^T y, A^T y \geq c, y \geq 0$. Derived using $c^T x \leq y^T Ax \leq y^T b$.

Primal to Dual LP Flip objective and constraints, then transpose weight matrix A .

Duality Weak duality says $\max(\text{primal}) \leq \min(\text{dual})$ while strong says equality.

Independence, Matchings and Covers

Matching Subset of edges where no two edges share a common endpoint. A perfect matching covers all vertices. A maximal matching cannot be enlarged by adding an edge. A maximum matching has maximum possible size. Not all maximal matchings are maximum. The matching number $\nu(G)$ is the maximum size of a matching in G .

Vertex Cover Set of vertices that contains at least one endpoint of every edge.

The vertex cover number $\tau(G)$ is the minimum size of a vertex cover of G .

Proposition $\tau(G) \geq \nu(G)$. For bipartite graphs, equality holds.

Clique Subset of vertices that induces a complete subgraph. The clique number $\omega(G)$ is the size of the largest clique in G .

Independence Set Set of vertices S where no two vertices in S are adjacent. The independence number $\alpha(G)$ is the maximum size of an independent set in G .

Cliques and Independence $\alpha(G) = \omega(\bar{G}), \omega(G) = \alpha(\bar{G})$.

Kneser Graph $KG(n, k)$ is the graph where vertices are k -subsets of $[n]$ and two vertices are adjacent if their sets are disjoint. $\alpha(KG(n, k)) = \binom{n-1}{k-1}$.

Independence and Vertex Cover For n vertices, $\alpha(G) + \tau(G) = n$.

Line Graph Vertices are edges of G , while two vertices of $L(G)$ are adjacent if the corresponding edges in G share an endpoint. Then $\nu(G) = \alpha(L(G))$.

Dual LPs Let B be the vertex-edge incidence matrix. Then the primal LP is the fractional matching problem while the dual LP is the fractional vertex cover problem. Weak duality implies $\sum x_e \leq \sum y_v$, which gives us $\nu(G) \leq \tau(G)$.

Bipartite Matching

Bipartite Graphs Graph where $V = A \sqcup B$ (disjoint union). Equivalently, G has no odd cycle or $\chi(G) \leq 2$.

Perfect Matching A matching is perfect to A if every vertex of A is matched.

$|A| \leq |B|$ is necessary.

Hall's Marriage Theorem A bipartite graph contains a matching perfect to A iff the marriage condition holds; $\forall S \subseteq A, |N(S)| \geq |S|$. Fails on infinite graphs.

Edge Decomposition The edges of a d -regular bipartite graph can be decomposed into d perfect matchings; repeatedly remove a perfect matching to get a $d-1$ -regular graph.

König's Theorem For any bipartite graph, $\nu(G) = \tau(G)$.

General Matching

Alternating Path (wrt M) Path such that v_1 is unmatched, and the edges alternate between not in M and in M . An augmenting path also ends at an unmatched vertex. Inverting an augmenting path increases the matching size.

Berge's Theorem A matching M is maximum iff there exists no augmenting path with respect to M .

Odd Component There are $o(H)$ components with an odd number of vertices.

Tutte's Theorem A graph has a perfect matching iff $\forall U \subseteq V, o(G[V \setminus U]) \leq |U|$.

Tutte-Berge Formula $\nu(G) = \frac{1}{2} \min_{U \subseteq V} (|U| - o(G[V \setminus U]) + n)$; alternatively, $2\nu(G) = n - \max_{U \subseteq V} (o(G[V \setminus U]) - |U|)$.

Deficiency $(U) = o(G[V \setminus U]) - |U|$. If positive, then at least (U) vertices must remain unmatched since we cannot fully match odd components.

Matching ILP Introduce a binary variable x_e for each edge. Then the constraints are $\sum_{v \in e} x_e \leq 1$ and we want to maximise $\sum_{e \in E} x_e$.

Berge's and LP An augmenting path is a feasible direction increasing the objective value. Berge's states local optimality (no augmenting path) implies global optimality.

Tutte's and LP Tutte's inequalities characterise when a solution exists for a perfect matching.

Tutte-Berge as Duality The primal problem is the maximum matching size, while the dual problem is maximising the deficiency. Tutte-Berge expresses the strong duality of these problems.

Connectivity

Separating Set / Vertex Cut A set $S \subset V(G)$ such that $G - S$ has more than one connected component.

Cut Vertex Separating set consisting of a single vertex.

Vertex Connectivity $\kappa(G)$, minimum size of $S \subset V(G)$ such that $G - S$ is disconnected or has only one vertex.

k-Connected $\kappa(G) \geq k$. Removing $k-1$ vertices leaves the graph connected.

2-Connected Graphs Every 2-connected graph has a cycle.

Connectivity and Minimum Degree $\kappa(G) \leq \delta(G)$.

Mader's Theorem Every graph of average degree at least $4k$ contains a k -connected subgraph.

Edge Connectivity A graph is ℓ -edge-connected if for every subset of edges with $|F| < \ell$, $G - F$ remains connected. $\kappa'(G)$ is the maximum such ℓ .

Edge Cuts and Bridges Every minimal disconnecting set is an edge cut. Any edge whose removal disconnects G is a cut edge or bridge. $\kappa'(G) = 1$ iff G has a bridge.

Cut Relations $\kappa(G) \leq \kappa'(G) \leq \delta(G)$.

Chartrand-Harary Theorem For all $0 < a \leq b \leq c$, there exists G such that $\kappa(G) = a, \kappa'(G) = b, \delta(G) = c$.

Whitney's Theorem A graph is 2-connected iff for every pair of vertices there exists two internally-disjoint $u - v$ paths; they belong to a cycle.

A-B Path Path such that $V(P) \cap A = \{x_0\}, V(P) \cap B = \{x_k\}$.

Separating Sets A set separates A and B if every $A - B$ path contains a vertex or edge from X .

Cut Vertex Vertex separating two other vertices in the same component.

Cut Edge Edge separating its two endpoints.

Menger's Theorem The minimum number of vertices separating A from B is equal to the maximum number of vertex-disjoint $A-B$ paths.

Corollaries of Menger's Theorem If a and b are not adjacent, then the minimum number of vertices separating them equals the maximum number of independent (internally vertex-disjoint) $a-b$ paths. The minimum number of edges separating a and b equals the maximum number of edge-disjoint $a-b$ paths.

Global Menger's Theorem A graph is k -vertex-connected iff every pair of vertices is joined by k internally-disjoint paths. A graph is k -edge-connected iff every pair of vertices is joined by k edge-disjoint paths.

Menger's as Duality If both the packing primal LP and covering dual LP are feasible, then Menger's is a strong duality theorem.

Flows

Flow Network Directed graph with a source (all out-edges) and sink (all in-edges), as well as a non-negative capacity on each edge. A flow respects capacity constraints at each node.

Size of Flow $|f| = \sum_{v:s \rightarrow v} f_{s \rightarrow v} = \sum_{v:v \rightarrow t} f_{v \rightarrow t}$.

Max-Flow LP Maximise flow under flow conservation constraints which respect capacities. Have a variable for each edge.

Cut Partition such that s and t fall into different components.

Capacity of Cut $c(S, T) = \sum_{u \in S, v \in T, u \rightarrow v \in E(D)} c_{u \rightarrow v}$, only forward edges.

Flow Across Cut The net flow going from nodes in S to nodes in T . The net flow across any cut is equal to the global flow. For every flow and cut, $|f| \leq c(S, T)$.

Min-Cut Cut minimising $c(S, T)$. $\max_f |f| \leq \min_{(S, T)} c(S, T)$.

Min-Cut LP Introduce real z_v for each vertex. Then minimise $\sum c_{u \rightarrow v} (z_v - z_u)$, subject to $z_s \leq 0, z_t \geq 1, z_v - z_u \geq 0$. z_v is the "side" of the cut containing v , with vertices with smaller z_v lying on the s -side.

Max-Flow Min-Cut Theorem The Max-Flow and Min-Cut LPs form a Primal-Dual Pair. $\max_f |f| = \min_{(S, T)} c(S, T)$.

Integral Flow Theorem If all capacities are integers, there exists a max-flow such that all edge flows are also integers.

Totally Unimodular Matrix Every square submatrix has determinant in $\{0, -1, 1\}$. They imply integrality of LP vertices when the RHS is integral. Bipartite / directed graphs have a TU incidence matrix.

Menger's (Directed, Edge-Disjoint)

- Consider a flow network where every edge has capacity 1.
- By Integral Flow Theorem, the max flow is an integer that can be decomposed into unit flows along edge-disjoint $x-y$ paths.
- A min-cut then has capacity equal to the number of edges that must be removed to disconnect the graph.
- Then by duality this is equal to the maximum number of edge-disjoint paths.
- For vertices, split them. For undirected, clone edges.

König's Theorem

- Construct the flow network where the source is connected to A and the sink to B with capacity 1. Then have edges from A to B with capacity ∞ .
- Max-Flow = $\nu(G)$: An integral flow sends 0 or 1 on each source or sink edge. The edges across the middle carrying flow 1 form a matching. Any matching gives a feasible integral flow.
- Min-Cut = $\tau(G)$: A finite cut cannot use any ∞ -edge. So the cut edges are a subset of the source or sink edges. This cuts all $s-t$ paths iff $A' \cup B'$ is a vertex cover. The capacity is $|A'| + |B'|$.
- Max-Flow Min-Cut gives equality.

Flow Numbers

Nowhere-Zero *k*-Flows Map $f : E(G) \rightarrow \{1, 2, ..., k - 1\}$ satisfying flow conservation. The graph must be bridgeless.

Flow Number $\varphi(G)$, smallest k for which a nowhere-zero k -flow exists.

2-Flow G has a 2-flow iff every vertex has an even degree.

3-Flow G has a 3-flow iff it is bipartite.

Complete Graph For all even $n > 4$, $\varphi(K_n) = 3$.

4-Flow Every 4-edge-connected graph has a 4-flow.

Theorem (Seymour's) Every bridgeless graph has a 6-flow, providing an upper bound.

Tutte's Conjectures

- 5-Flow: Every bridgeless multigraph has a 5-flow.
- 4-Flow: Every bridgeless multigraph with no Petersen-graph minor has a 4-flow.
- 3-Flow: Every multigraph with no cut of size 1 or 3 has a 3-flow.

Abelian Flow An A -flow is a function $f : E(G) \rightarrow A$ satisfying Kirchoff's law at every vertex; $\sum_{in} f(e) = \sum_{out} f(e)$. An example is $M = \mathbb{Z}_k = \mathbb{Z}/k\mathbb{Z}$ (modulo k).

Theorem (Tutte) Nowhere-zero k -flow exists iff nowhere zero $\mathbb{Z}_k$ -flow exists.

Colouring

Proper Colouring A proper k -colouring is a map $c : V(G) \rightarrow \{1, ..., k\}$ such that $c(u) \neq c(v), uv \in E(G)$. $\chi(G)$ is the smallest k for which a proper k -clouring exists. k -colourable means $\chi(G) \leq k$.

Theorem (Lovász) $\chi(KG(n, k)) = n - 2k + 2$.

Bounds $\chi(G) \geq \omega(G), \chi(G) \geq |V(G)|/\alpha(G)$.

$$\chi(G) \leq \frac{1}{2} + \sqrt{2m + \frac{1}{4}}, \chi(G) \leq \Delta(G) + 1.$$

Criticality G is k -critical if $\chi(G) = k$ but removing any edge or vertex decreases χ . Odd cycles are 3-critical. If H is k -critical, $\delta(H) \geq k - 1$.

$$\chi(G) \leq 1 + \max_{H \subseteq G} \delta(G).$$

Theorem (Brooks) Let G be a connected graph which is not complete or an odd cycle. Then $\chi(G) \leq \Delta(G)$.

Hajós Construction for k -constructible graphs

- K_k is k -constructible.
- If x, y are non-adjacent, then $(G + xy)/xy$ is k -constructible.
- If $V(G_1) \cap V(G_2) = \{x\}, xy_i \in E(G_i)$, then $(G_1 \cup G_2) - xy_1 - xy_2 + y_1y_2$ is k -constructible.
- $\chi \geq k$ iff G contains a k -constructible subgraph.
- Every k -critical graph is k -constructible.

Edge Colouring Map $c : E(G) \rightarrow S, c(e) \neq c(f)$ if they share a vertex. $\chi'(G)$ is the smallest k with proper k -edge colouring. $\chi'(G) = \chi(L(G))$, $\chi'(G) \geq \Delta(G)$.

König's Theorem Every bipartite graph satisfies $\chi'(G) = \Delta(G)$.

Vizing's Theorem Every graph satisfies $\Delta(G) \leq \chi'(G) \leq \Delta(G) + 1$ (class 1 or class 2). For any loopless multigraph, $\Delta(G) \leq \chi'(G) \leq \Delta(G) + \mu(G)$, $\mu(G)$ is the maximum edge multiplicity.

Shannon's Theorem Every multigraph satisfies $\chi'(G) \leq \lfloor \frac{3}{2} \Delta(G) \rfloor$.

Perfect Graph $\chi(H) = \omega(H)$ for every induced subgraph.

Planarity

Planar Graph Can be drawn such that edges are curves intersecting only at their endpoints. $K_5, K_{3,3}$ are non planar.

Plane Graph Planar embedding of a planar graph. Plane consists of connected regions called faces, the unbounded region is the outer face while the others are inner faces.

Euler's Formula For a connected plane graph with n vertices, m edges, and ℓ faces, $n - m + \ell = 2$. Every planar graph with $n \geq 3$ vertices has at most $3n - 6$ edges.

Fáry's Theorem Every planar graph has a straight-line drawing without crossings. Harborth's conjecture extends this to lines of integral lengths.

Crossing Number $cr(G)$ is the minimum number of edges crossings over all drawings of G in the plane. $cr(G) \geq m - 3n + 6$. If $m \geq 4n, cr(G) \geq \frac{m^3}{64n^2}$.

Zarankiewicz's Conjecture $cr(K_{m,n}) = \lfloor \frac{m}{2} \rfloor \lfloor \frac{m-1}{2} \rfloor \lfloor \frac{n}{2} \rfloor \lfloor \frac{n-1}{2} \rfloor$, upper bound can be constructed. For complete graphs $cr(K_n) \leq \frac{1}{4} \lfloor \frac{n}{2} \rfloor \lfloor \frac{n-1}{2} \rfloor \lfloor \frac{n-2}{2} \rfloor \lfloor \frac{n-3}{2} \rfloor$.

Subdivision A graph obtained by replacing each edge by a path of length ≥ 1 where these paths are internally vertex-disjoint.

Topological Minor $G \supseteq TH$ if G contains a subgraph isomorphic to a subdivision of H . The original vertices of H are branch vertices.

Kuratowski's Theorem G is planar iff it contains no subdivision of K_5 or $K_{3,3}$. **Minor** $G \supseteq MH$ if H can be obtained from G by a sequence of deleting edges or vertices, or contracting edges. Every topological minor of G is also a minor of G . If H has max degree at most 3, $G \supseteq MH$ implies $G \supseteq TH$.

Equivalence Lemma G contains K_5 or $K_{3,3}$ as a minor iff it contains them as a topological minor. This shows Wagner's (minor) is equivalent to Kuratowski's.

Hanani-Tutte Theorem Two edges are independent if they share no endpoint. G is planar iff it has a drawing in which every pair of independent edges crosses an even number of times.

Colouring Planar Graphs

Four Colour Theorem Every planar graph satisfies $\chi(G) \leq 4$.

Kempe Chain The Kempe (i, j) -chain from u is the connected component of u in the subgraph induced by all vertices with those colours.

Theorem (Grötzsch) Every triangle-free planar graph is 3-colourable.

Geometric Dual Graph Faces become vertices with new edges if they share a boundary, also planar.

Flows and Colourings Orient G and colour the faces with values in $\mathbb{Z}_k$. For each edge, take the difference of face colours. A nowhere-zero $\mathbb{Z}_k$ flow on G determines a proper colouring of G^* .

Planar Duality G^* has a proper $\mathbb{Z}_k$ -colouring $\Leftrightarrow G$ has a nowhere-zero $\mathbb{Z}_k$ -flow. To upgrade to integer flows, use Tutte's.

Polynomials

Chromatic Polynomial $P(G, k)$ is the number of proper k -colourings of G . For n vertices and m edges, let $P(G, k) = \sum_{i=0}^n a_i k^i$. Then $\deg P(G, k) = n, a_n = 1, a_{n-1} = -m, a_0 = P(G, 0) = 0$. The coefficients are all nonzero and alternate in sign.

Deletion-Contraction $P(G, k) = P(G - uv, k) - P(G/uv, k)$. Either the 2 endpoints have the same colour or they do not.

$$P(C_n, k) = (k - 1)^n + (-1)^n (k - 1).$$

Read's Conjecture The absolute values $|a_n|, ..., |a_1|$ are unimodal.

Huh's Theorem The sequence $|a_i|$ is log-concave, $a_i^2 \geq a_{i-1} a_{i+1}$, and is unimodal.

Chromatic Roots For an n vertex graph, $P(G, k)$ has no real root greater than $n - 1$, no negative real roots, and no roots in $(0, 1)$.

Domain No real roots lie in $(1, \frac{32}{27}]$, they are dense in $[\frac{32}{27}, \infty)$; the bound is sharp. Complex chromatic roots are dense in $\mathbb{C}$.

Acyclic Orientations Choice of direction of each edge such that no directed cycle is created.

Theorem (Stanley) $a(G) = (-1)^n P(G, -1)$.

Flow Polynomial $F(G, k)$ counts the number of nowhere-zero $\mathbb{Z}_k$ flows on G .

Proposition If A, A' are finite with same cardinality, then G has the same number of nowhere-zero flows.

Deletion-Contraction for Flows Choose an edge which is neither a bridge nor a loop, then $F(G, k) = F(G/e, k) - F(G - e, k)$. For a loop, $F(G, k) = (k - 1)F(G - e, k)$. For a bridge, $F(G, k) = 0$.

Planar Duality $F(G, k) = \frac{1}{k} P(G^*, k)$. Free choice of colour on one face.

Tutte Polynomial $T_G(x, y) = \sum_{A \subseteq E} (x - 1)^{r(E) - r(A)} (y - 1)^{|A| - r(A)}$ where $r(A) = |V| - c(A)$ is the rank of A , $c(A)$ is the number of connected

components in the spanning subgraph (V, A) . The first exponent is corank, how many more components A has than G . The second exponent is nullity, the dimension to the cycle space of (V, A) , the number of independent cycles. T_G is a generating function over all subgraphs.

Deletion-Contraction for Tutte Polynomial If e is ordinary, $T_G = T_{G-e} + T_{G/e}$. If e is a bridge, $T_G = x \cdot T_{G/e}$. If e is a loop, $T_G = y \cdot T_{G-e}$.

Specialisation For $n = |V|, m = |E|, c = c(G)$, $P(G, k) = (-1)^{n-c} k^c T_G(1 - k, 0), F(G, k) = (-1)^{m-n+c} T_G(0, 1 - k)$.

Evaluations of $T_G(x, y)$: $T_G(1, 1)$: Spanning trees. $T_G(2, 1)$: Spanning forests. $T_G(1, 2)$: Connected spanning subgraphs. $T_G(2, 0)$: Acyclic orientations. $T_G(2, 2)$: Edge subsets.

Notation

- $\delta(G)$: Minimum Degree, $\Delta(G)$: Maximum Degree
 - $\nu(G)$: Matching number / Maximum matching size
 - $\tau(G)$: Vertex cover number / Minimum vertex cover
 - $\omega(G)$: Clique number / Maximum clique size
 - $\alpha(G)$: Independence number / Maximum independent set size
 - $o(H)$: Number of odd components
 - $\kappa(G)$: Vertex connectivity / Minimum size of vertex cut
 - $\kappa'(G)$: Edge connectivity / Minimum size of edge cut
 - $\varphi(G)$: Flow number / Smallest k with nowhere-zero k flow
 - $\chi(G)$: Chromatic number / Smallest k with k -colouring.
 - $\chi'(G)$: Chromatic index / Smallest k with k -edge-colouring.
 - $cr(G)$: Crossing number / Minimum number of edge crossings.
 - $P(G, k)$: Chromatic Polynomial / Proper k -colourings of G .
 - $a(G)$: Acyclic orientations of G .
 - $F(G, k)$: Flow Polynomial / Nowhere-zero $\mathbb{Z}_k$ flows of G .
- Petersen Graph** $\nu(P) = 5, \tau(P) = 6, \alpha(P) = 4, KG(5, 2), \kappa(P) = 3, \kappa'(P) = 3, \chi(P) = 3, \chi'(P) = 4, cr(P) = 2, \varphi(P) = 5$.

Tutorial Stuff

- T1Q2** If G is regular, then so is $L(G)$. G is connected iff $L(G)$ is connected.
- T1Q3** Girth is the length of the shortest cycle in the graph. A k -regular graph with girth 5 has at least $k^2 + 1$ vertices.
- T2Q1** $\alpha(G) \geq \frac{|V(G)|}{\Delta(G)+1}, \tau(G) \leq |V(G)| \frac{\Delta(G)}{\Delta(G)+1}$.
- T3Q2** Every 3-regular graph with no cut edge has a perfect matching.
- T4Q1** For $k \geq 2$, a connected k -regular bipartite graph is 2-connected.
- T4Q2** If a graph is 3-regular, then $\kappa(G) = \kappa'(G)$.
- T4Q4** G is 2-connected iff for any x, y, z , there is a path from x to y containing z .
- T5Q1** The signed incidence matrix for a directed graph is TU.
- T6Q1** For two graphs sharing a vertex set, $\chi(G_1 \cup G_2) \leq \chi(G_1) \cdot \chi(G_2)$.
- T6Q2** The tensor product has vertex set $V(G) \times V(H)$ where $(g_1, h_1), (g_2, h_2)$ are adjacent iff $g_1 \sim g_2, h_1 \sim h_2$. $\chi(G \times H) \leq \min\{\chi(G), \chi(H)\}$. Equality holds for $\chi(G) = \chi(H) = 3$.
- T7Q2** $2\sqrt{n} \leq \chi(G) + \chi(\overline{G}) \leq n + 1, n \leq \chi(G) \cdot \chi(\overline{G}) \leq (n + 1)^2/4$.
- T7Q3 Art Gallery Theorem** For a polygon of n walls, at most $\lfloor \frac{n}{3} \rfloor$ guards are needed.

- T8Q1** There exists k -regular planar graphs only for $k \leq 5$.
- T8Q2** There exists a 8 vertex graph where $G, \overline{G}$ are planar. There does not exist 11 vertex graph such that $G, \overline{G}$ are planar.
- T9Q1** Every graph admits an embedding in $\mathbb{R}^3$ in which each edge is drawn as a straight line segment.
- T9Q2** Every planar graph is a union of three forests.
- T9Q3** Every minor of a planar graph is planar.
- T9Q5** If G is 2 or 3-connected, then G^* also is. This does not hold for 4-connectedness.